Wabash Capstone Project Notes 06

By: John Danison

What Did I Say I Would Do This Week?

- Perform Physical Calibration.
- Begin writing user interface software for data offloading.

What I Have Done Over The Past Week

- Updated code to make new mode for system calibration testing.
- Performed system calibration testing.
- Began Wabash Data Offloading Software (Wabash Receiver Command Center).
- Attended the meetings with the Wabash team to get additional feedback & software functionality.

Updated Code w/ New Mode

I made another line option “L” to allow for continuous data capture and then to write this to the SD card.

```
--- Serial Commands ---
  s - Sync time via WiFi (requires WiFi credentials in main.h)
  t - Display current time
  d - Display all stored events
  c - Clear all events from SD card
  o - Offload data (playback events, resync time, clear SD)
  g - Read single strain gauge sample
  z - Tare/zero the strain gauge
  r - Restart NAU7802 conversions (if timeouts occur)
  m - Monitor strain continuously (press any key to stop)
  l - Lab test: Log strain readings to SD card (press any key to stop)
  b - Bridge balance and sensitivity test
  1-4 - Test with gain 1x, 2x, 4x, 8x (temporary)
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```

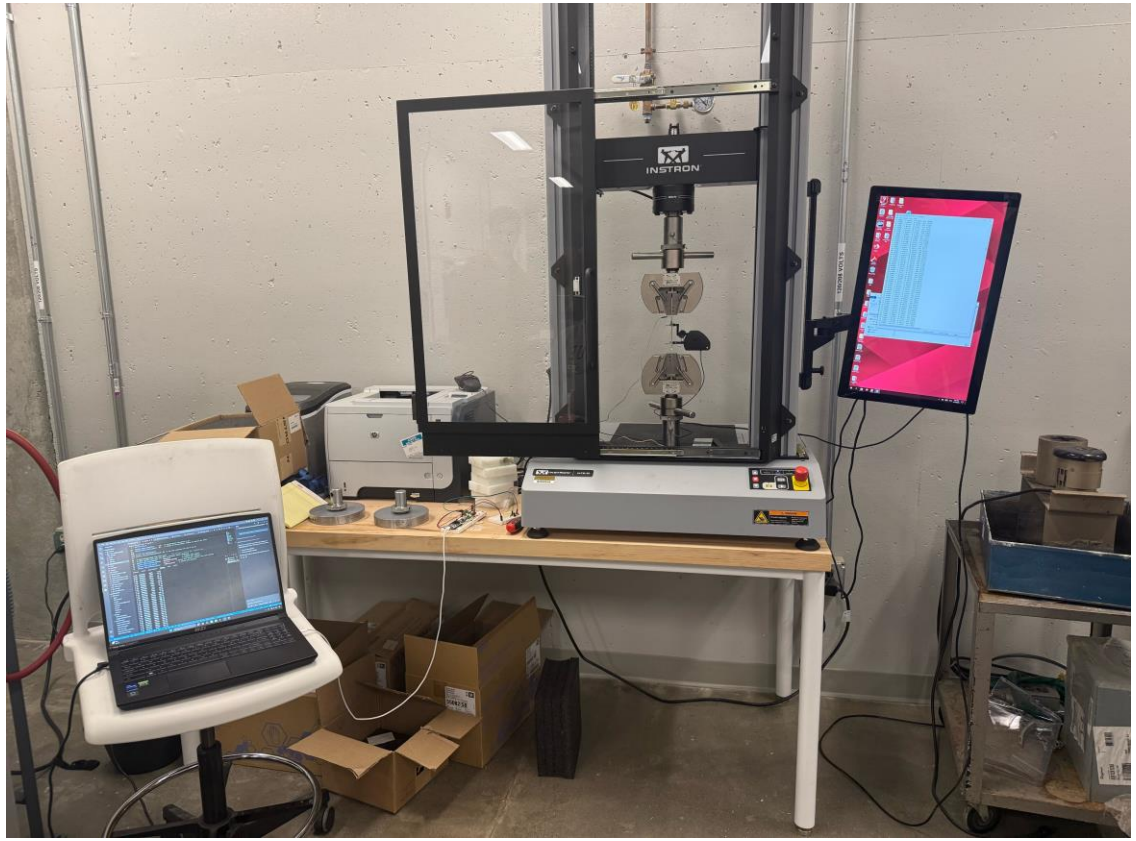
```
=== LAB TEST: CONTINUOUS STRAIN LOGGING ===
Sample Rate: 20 Hz
[LOG_START]
Recording raw strain gauge data...
Apply load now. Press any key to stop.

Time(s), Raw, Zeroed, Strain(με)
--------------------------------------
0.00, -8387527,        1,      0.00
0.06, -8387530,       -2,     -0.00
0.11, -8387527,        1,      0.00
0.16, -8387528,        0,      0.00
0.22, -8387529,       -1,     -0.00
0.27, -8387528,        0,      0.00
0.47, -8387527,        1,      0.00
0.52, -8387528,        0,      0.00
0.58, -8387527,        1,      0.00
0.63, -8387530,       -2,     -0.00
0.68, -8387527,        1,      0.00
0.74, -8387530,       -2,     -0.00
0.79, -8387527,        1,      0.00
0.84, -8387528,        0,      0.00
--------------------------------------
Monitoring stopped. Collected 14 samples.

Saving to SD card...
Data saved to: /lab-testing/strain-log12.txt
[LOG_END]
==========================
```

Calibration Testing - Setup

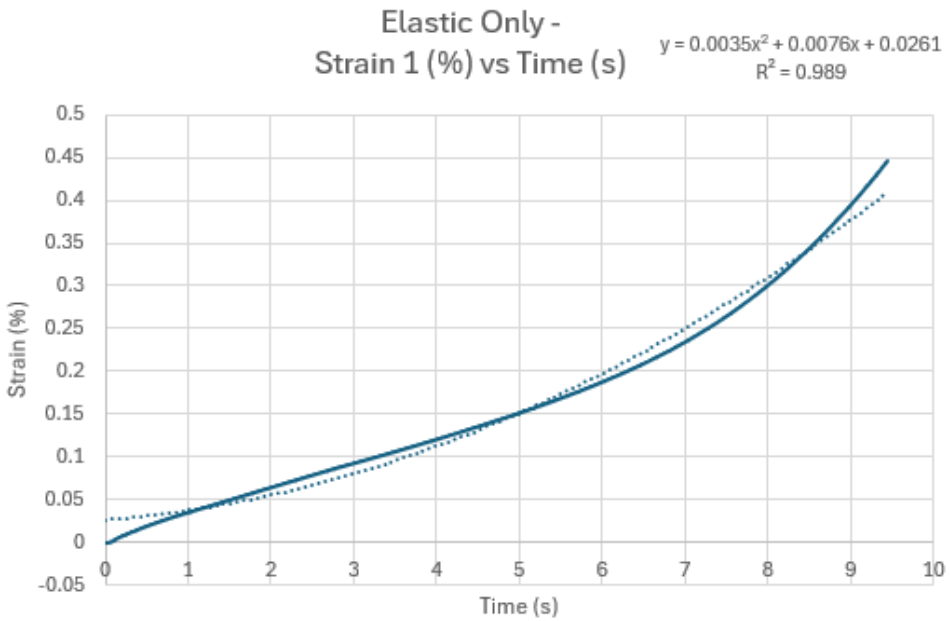
Extensometer with Brass Dog bone



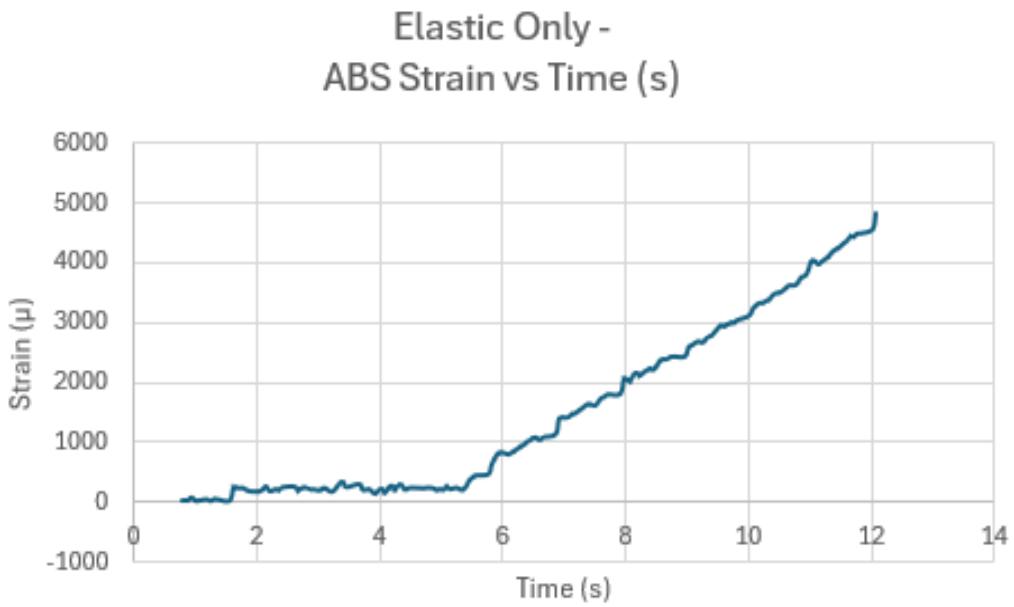
Calibration Testing - Results

- During Peak Strain, 100x difference.
- Numerically, values were within 3%
- Looking into a calibration formula

Extensometer:

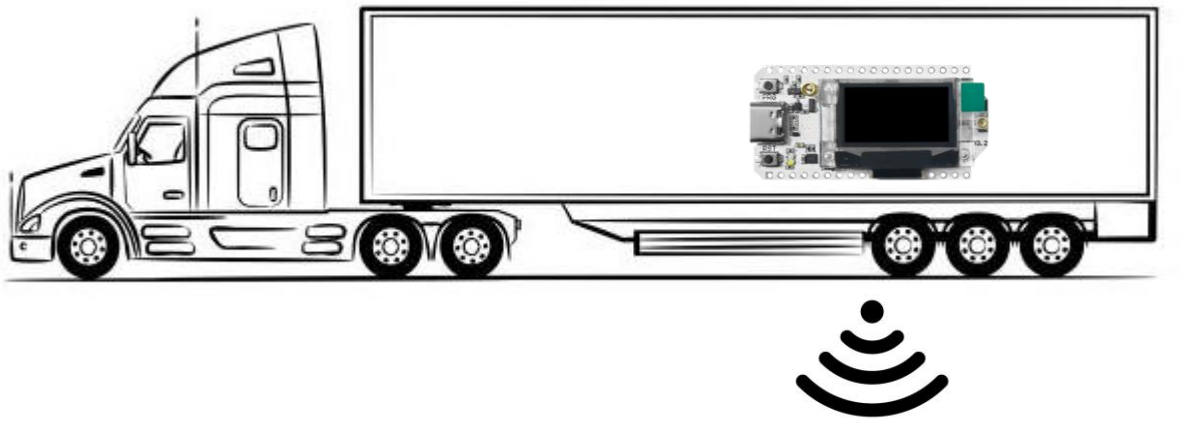


Our System:

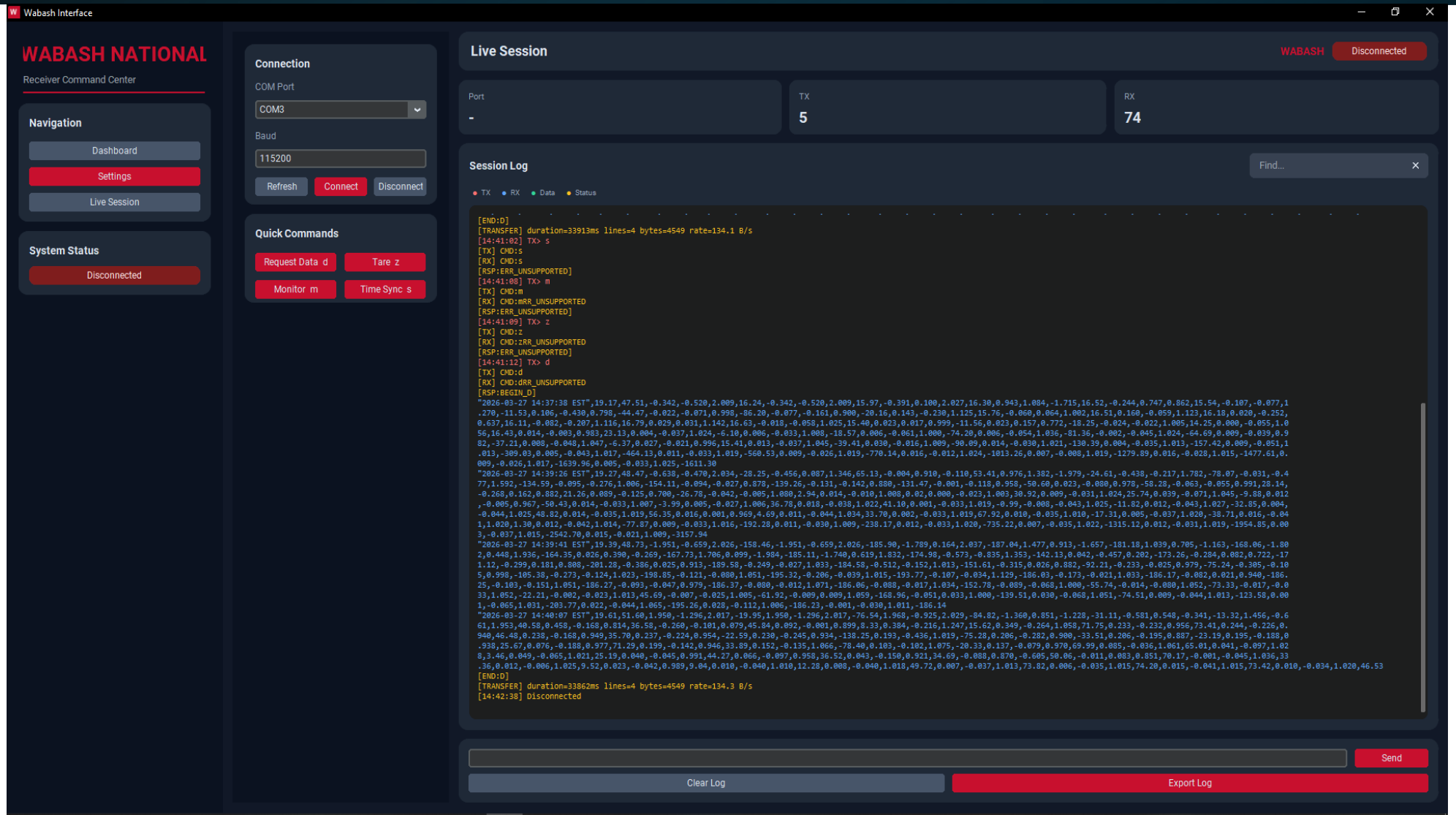


Wabash Data Offloading Software

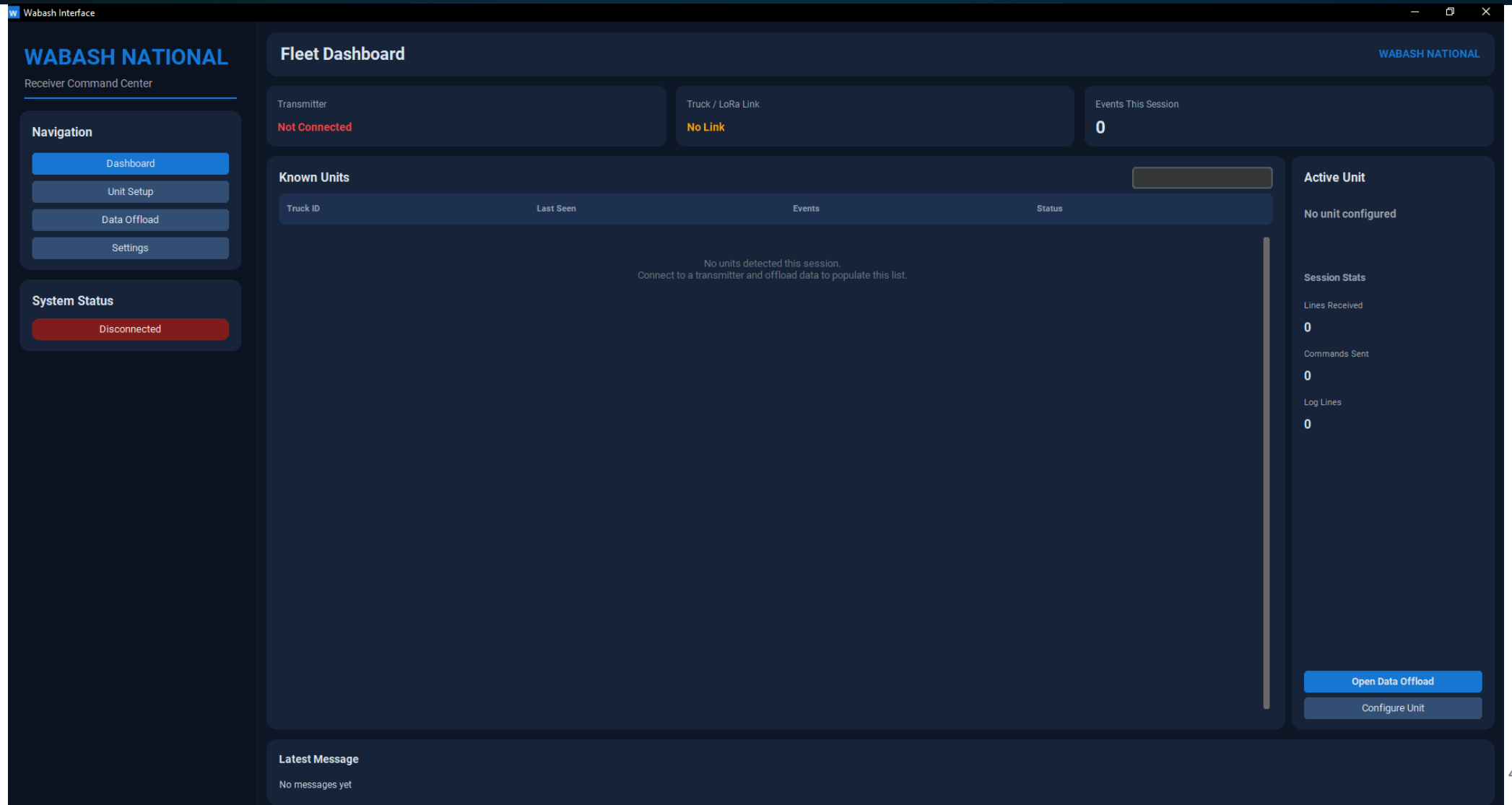
- **customtkinter** (main desktop UI framework)
- **pyserial** (serial communication with hardware)
- **pillow** (image handling for the UI/assets)
- **pyinstaller** (packaging the app into an executable)
- **setuptools** (build/packaging backend)
- **wheel** (build artifact support)



Wabash Data Offloading Software - Old



Wabash Data Offloading Software - Current



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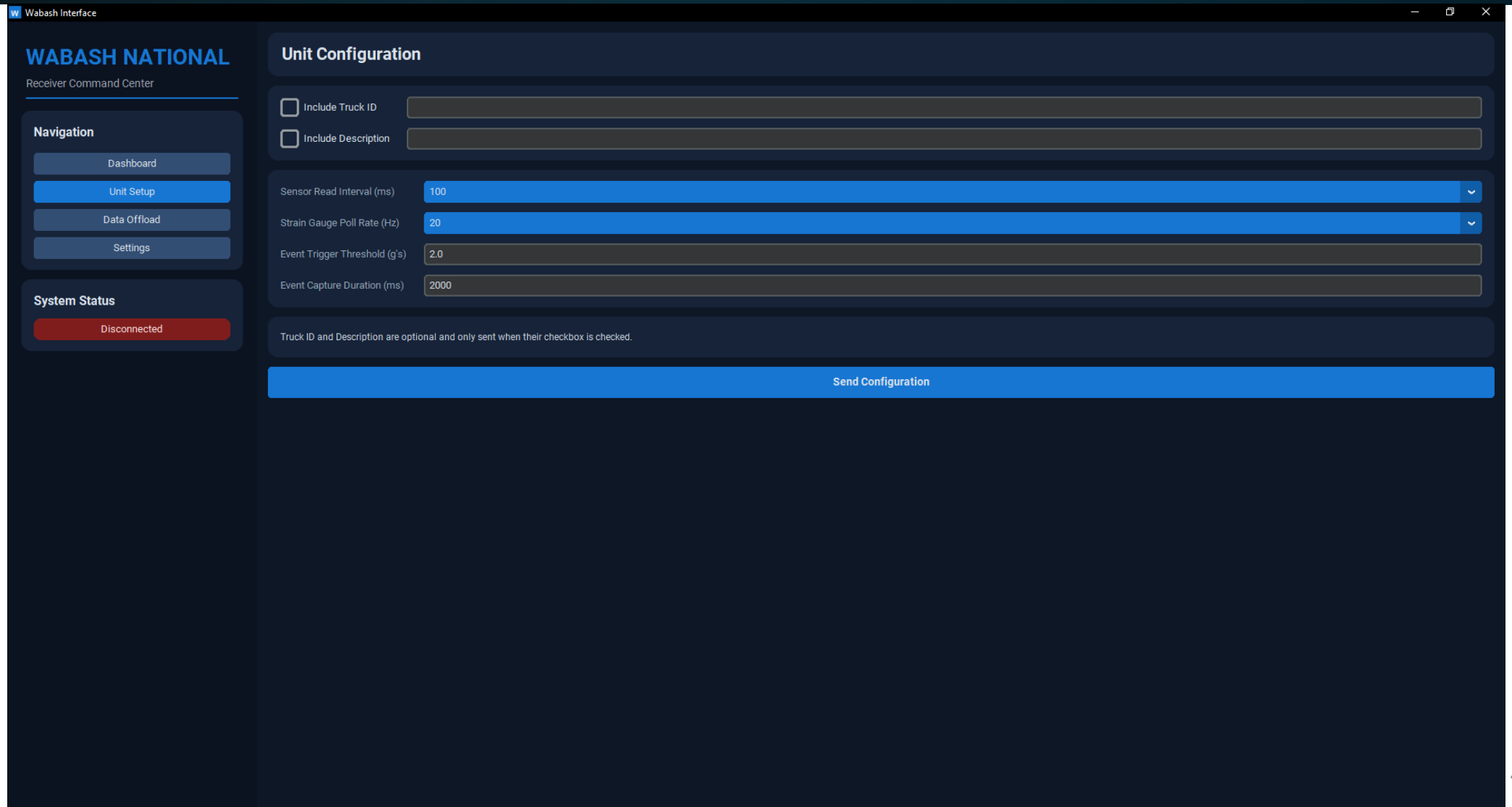
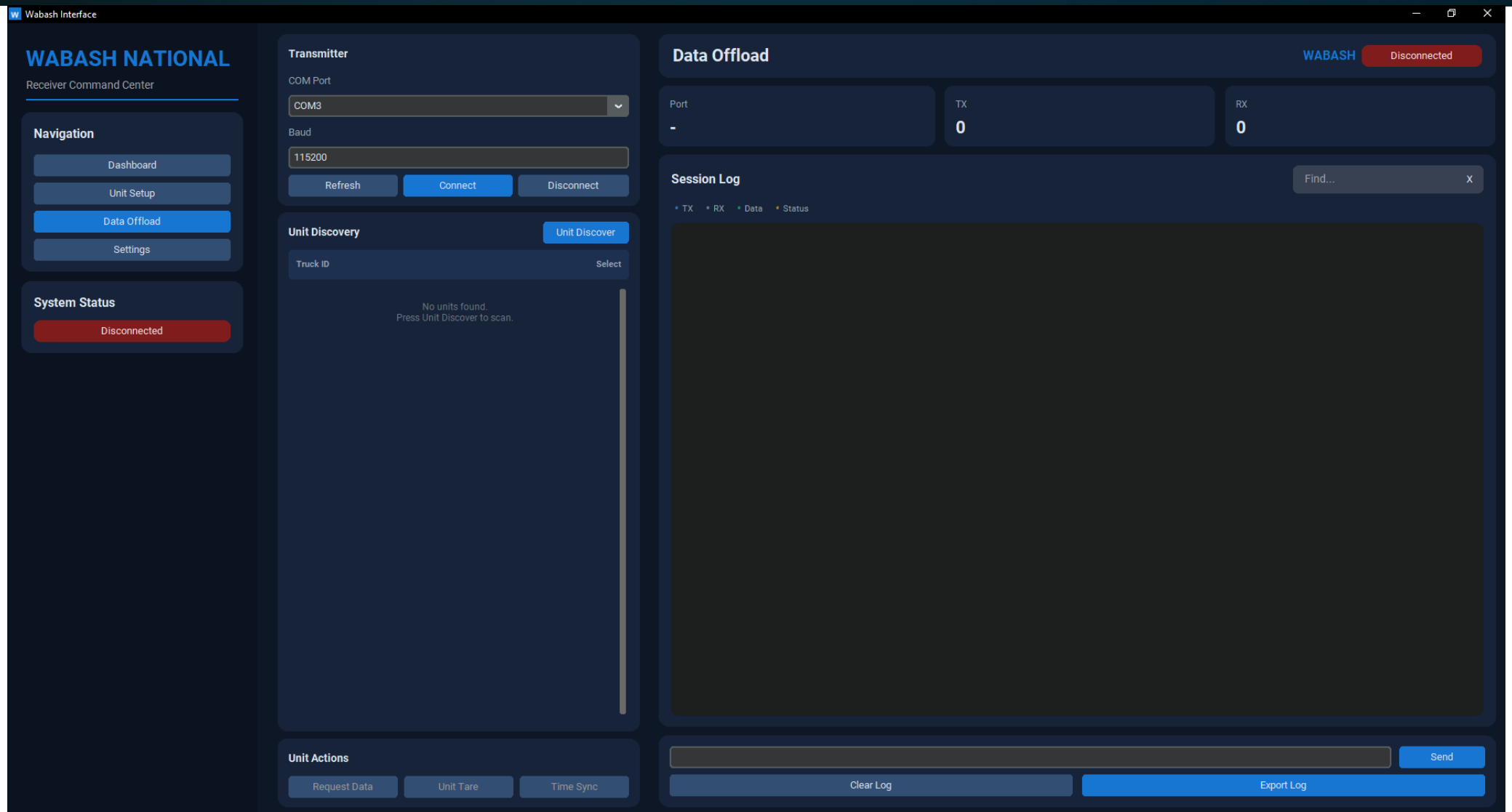
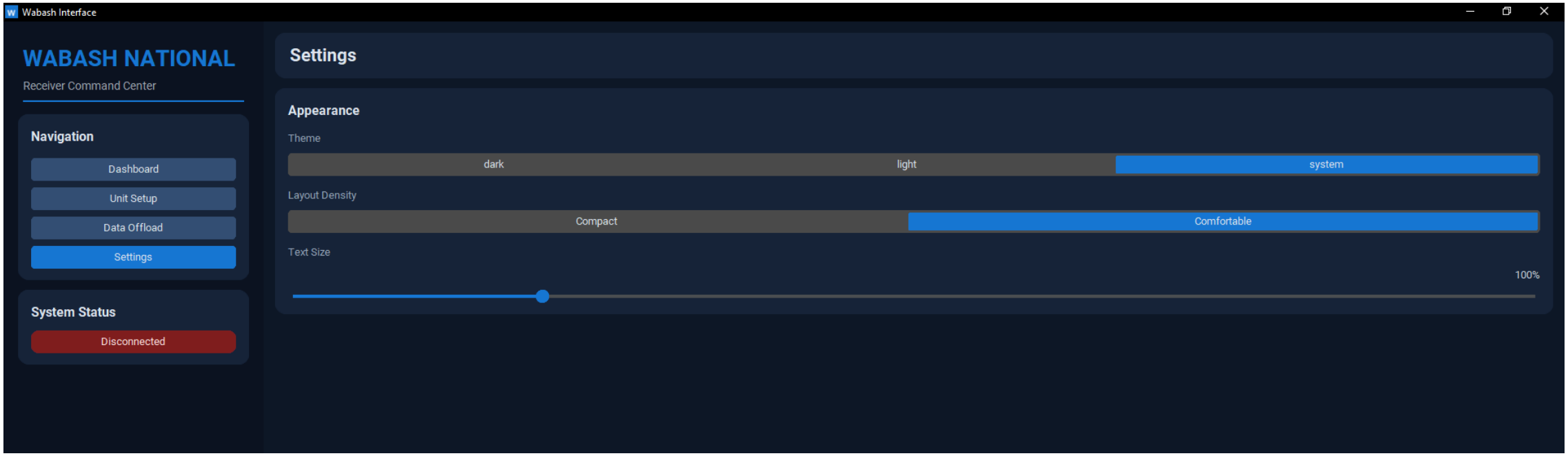


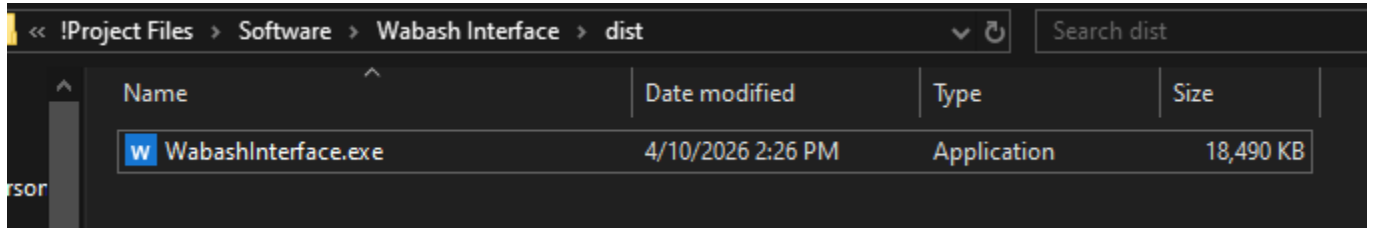
Wabash Data Offloading Software - Current



Wabash Data Offloading Software - Current



Packaged & Built .exe application



Did I Do What I Say I Would Do?

I feel that I accomplished everything that I said I would do over these past couple weeks.

The data offloading software is well under development and going well as of today.

I attended all lectures, labs, and meetings w/ the client.

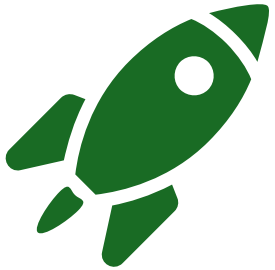
Plans For Upcoming Weeks

High Level

Continue to develop data offloading software.

Continue to show up to class.

Continue to support lab testing as requested by the team.



By EOW:

Continue to add client feature requests in data offloading software.